

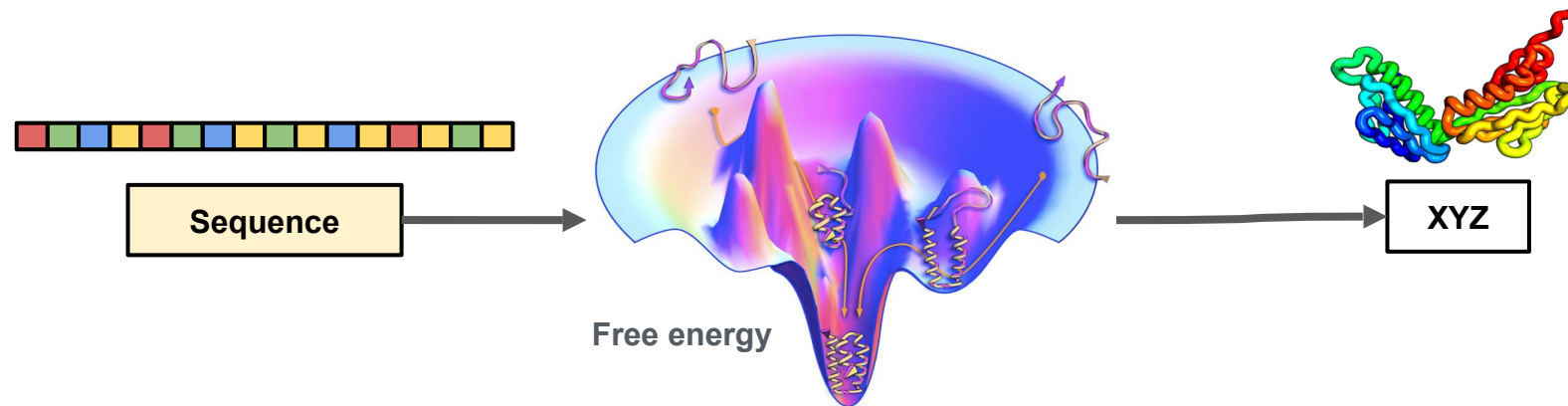
AlphaFold3, and how to read its confidence

A short introduction before Part 1

Cilia Connect 2026 · AI Structure Prediction Workshop · Esben Lorentzen

From sequence to structure

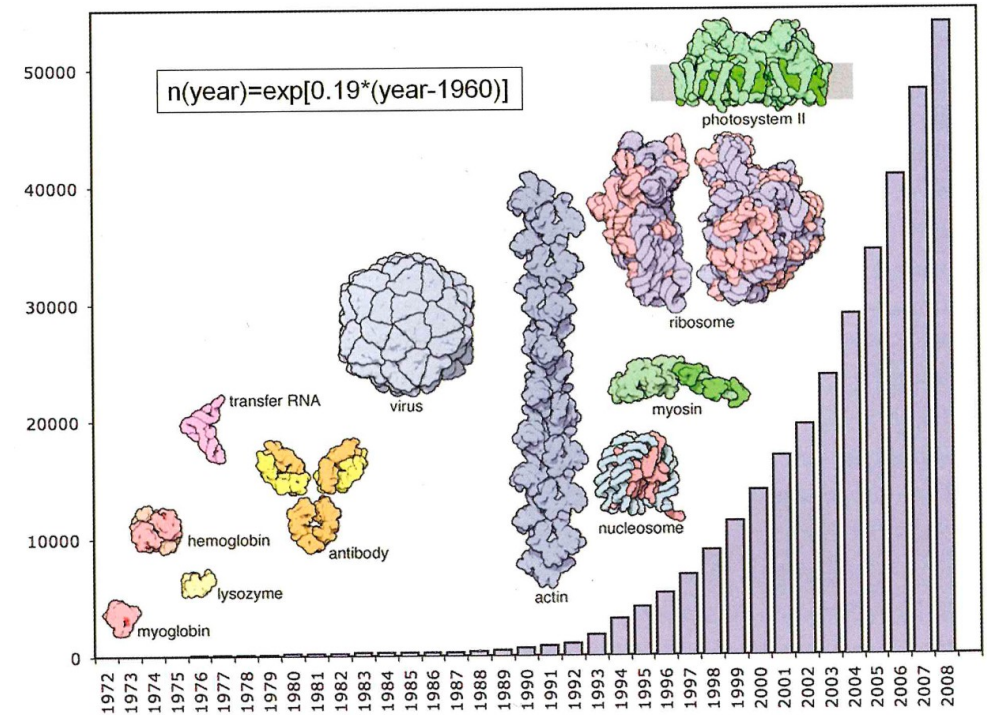
- A protein's amino-acid sequence encodes its 3D fold (Anfinsen, 1950s).
- That fold is simply the lowest free-energy state: the bottom of a folding funnel.
- So in principle the structure is predictable from sequence alone. In practice it is not: computing the fold from energy means searching an astronomical number of shapes.
- Solving structures experimentally works, but it is slow: only ~17% of human proteome residues have one.
- AlphaFold sidesteps the energy calculation entirely and predicts the fold directly from sequence, in minutes.



Funnel Image from: "The protein-folding problem, 50 years on." *science* 338, no. 6110 (2012): 1042-1046.

AlphaFold: a deep neural network trained on known structures

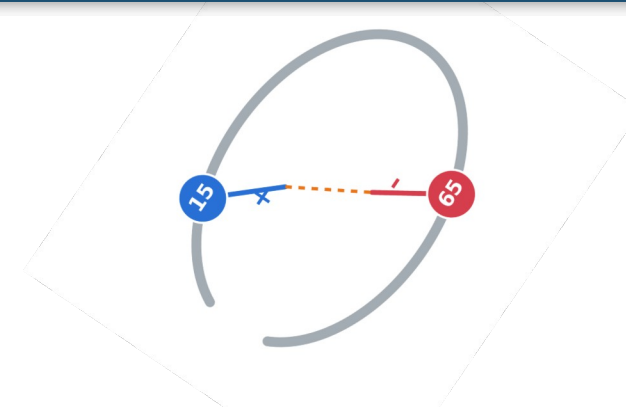
- AlphaFold is a deep-learning method: a large neural network, not a physics simulation.
- It was trained on the whole Protein Data Bank, 200,000+ structures from 60 years of crystallography, cryo-EM and NMR.
- From your sequence, plus an alignment of related sequences, it predicts the 3D coordinates directly.
- It learned the patterns that link sequence to shape, and it also reports how confident it is.



Growth of the Protein Data Bank since the 1970s.

Co-evolution reveals what is close in 3D

- Huge databases of sequences from many species are freely available.
- Orthologous proteins keep a similar fold even when their sequences have drifted far apart.
- So within an alignment, positions that touch in 3D tend to mutate in a coordinated way: co-evolution.
- Example: when position 15 changes charge, position 65 changes to match,
- Residues far apart in the sequence can therefore be close in 3D, and the alignment itself carries the 3D information that AlphaFold reads.



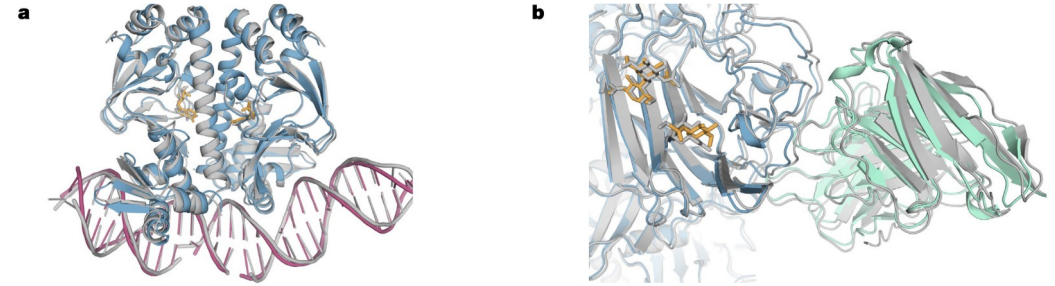
Multiple Sequence Alignment (MSA)

	15	20	35	50	65
Human	K	L	G	V	E
Mouse	R	I	G	L	D
Fish	K	V	G	I	E
Fly	E	M	G	F	K
Yeast	D	F	G	A	R

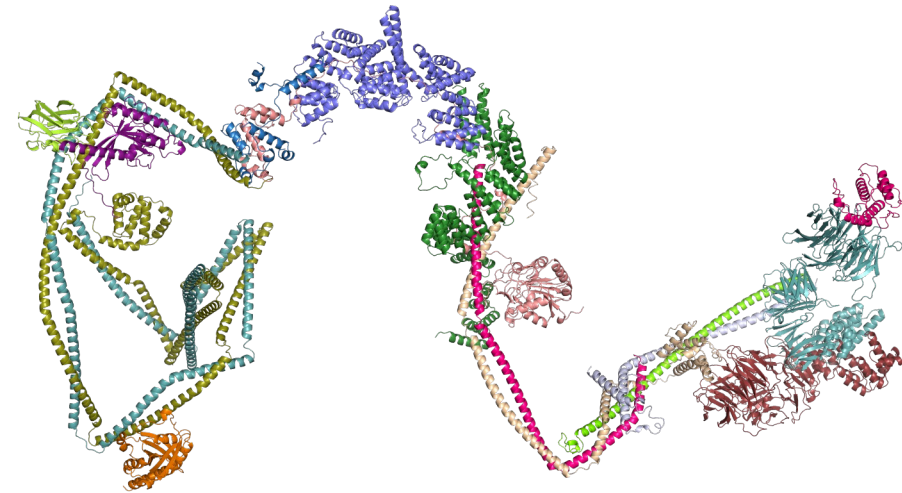
← Co-evolution: coordinated mutations →

What AlphaFold3 can predict

- Single proteins (monomers).
- Protein-protein complexes and large assemblies.
- Proteins together with ligands and metal ions.
 - For example your ARL13B with GTP and Mg^{2+} .
- Proteins with DNA and RNA.



Protein with DNA and ligands (Abramson et al., Nature 2024);



*Structure prediction of the IFT B complex.
Petriman et al., EMBO J. 2022)*

AlphaFold is not always correct but it grades its own work

Two different questions, two different scores:

- **pLDDT = predicted Local Distance Difference Test** (the "p" is predicted; IDDT is a per-residue measure of local structural accuracy).
per residue (0-100)

Is each residue's local structure right?

- **pTM = predicted TM-score** (TM = Template Modeling score, a global measure of how correct the overall fold is; 0–1).
whole model (0-1)

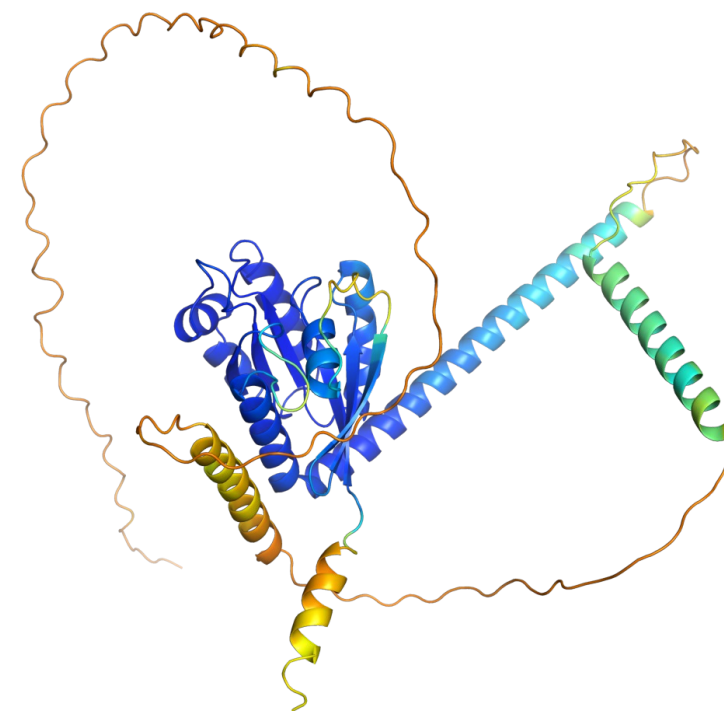
Is the overall fold and arrangement right?

A third score, PAE, describes relative positions and interfaces. We meet it in Part 2.

pLDDT: per-residue confidence

- One value per residue, 0 to 100. The 3D model is coloured by it.
- High = the local geometry is reliable. Low = flexible or uncertain.

	very high >90
	confident 70-90
	low 50-70
	very low <50



ARL13B (your example): blue folded G-domain core; orange/yellow disordered termini.

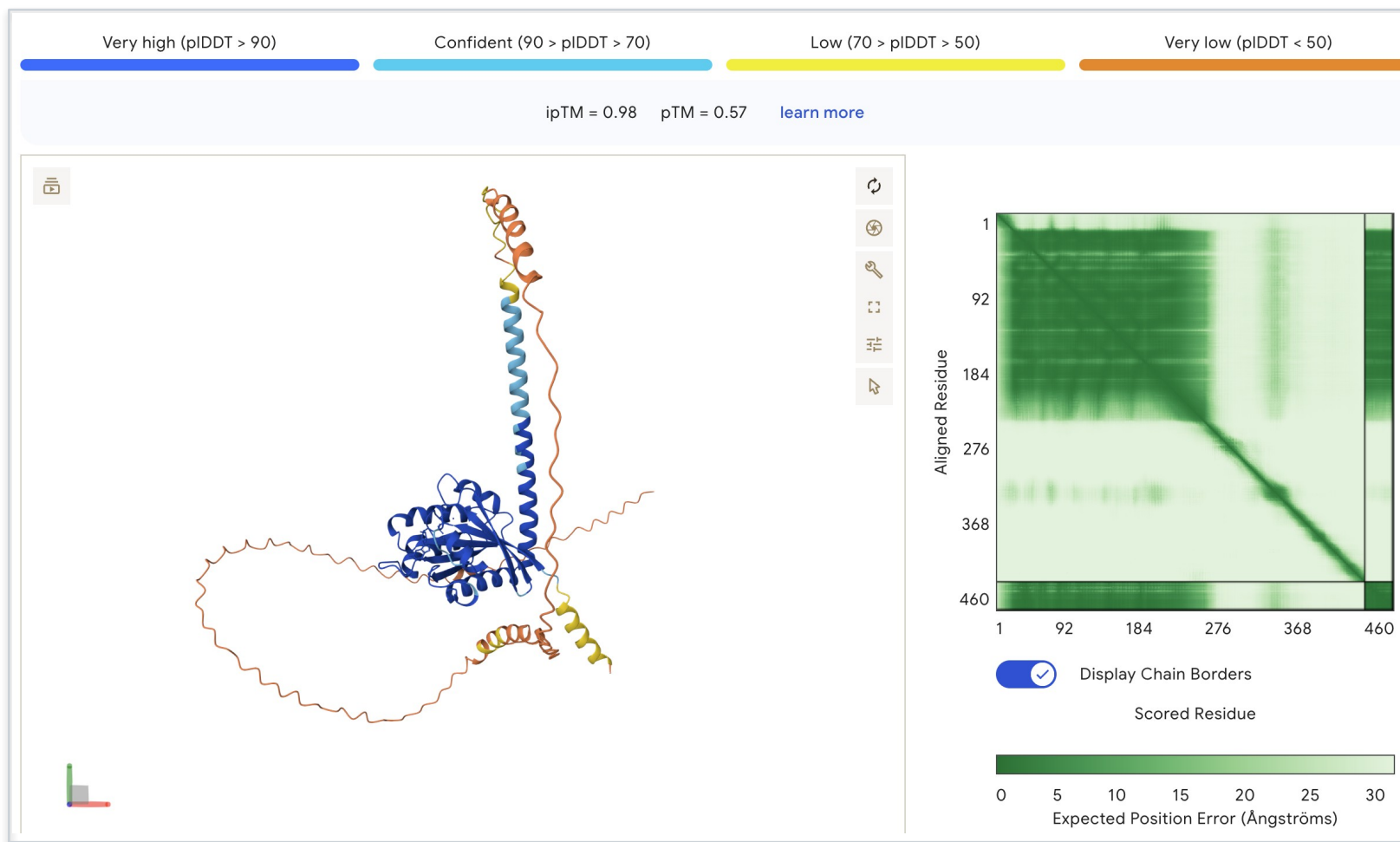
pTM: one global score

- A single number, 0 to 1, for the whole model's fold and domain arrangement.
- Rough guide: above ~0.5 the overall fold is probably about right.
- The models AlphaFold returns are ranked mostly by this kind of score.

Watch out: one number averages the whole protein.

A protein that is half well-folded and half disordered can score only modestly, even when its folded core is excellent. Always read pLDDT alongside pTM.

Where to find these on alphafoldserver.com

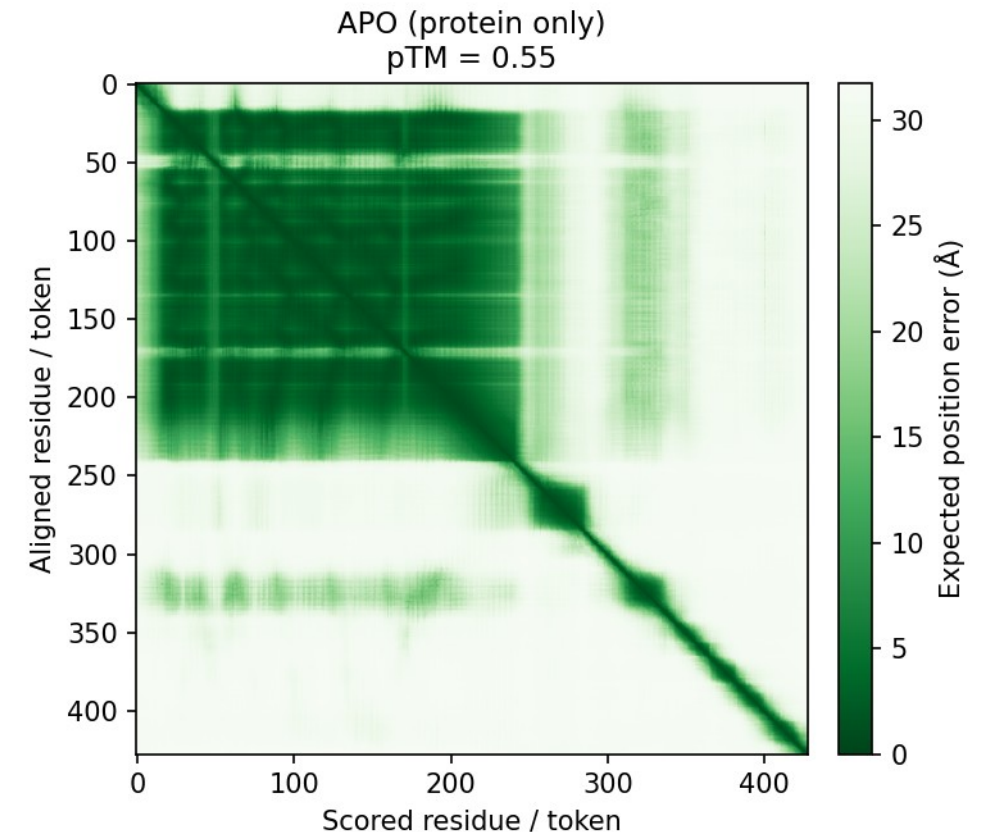


On the results page:

- 5 ranked models
- 3D viewer, coloured by pLDDT
- the pTM / ipTM scores
- a PAE tab (Part 2)
- download button for the 5 models

Next: relative positions and complexes

- pLDDT and pTM tell you about one model on its own.
- PAE tells you how confident AlphaFold is about the position of one part relative to another.
- In Part 2 we use PAE and ipTM to judge whether a predicted complex interface is real.



PAE plot: dark = confident relative position, light = unknown.